ADR IDENTIFICATION BASICS

Pharmacovigilance Internship – Task 1 | Educational Case Analysis

Important note
The patient scenarios in this report are fictional, created only for learning and practice. They are not real patient records and must not be treated as clinical advice.

1. Objective

To practice identifying suspected adverse drug reactions (ADRs) by reviewing a patient's symptoms, medication history, time relationship between medicine use and symptom onset, other possible causes, and the outcome after the suspected medicine is stopped or changed by a healthcare professional.

2. Basic Concepts

Pharmacovigilance is the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or other medicine-related problems. An **adverse drug reaction (ADR)** is a harmful and unintended response suspected to be associated with a medicine used at normally recommended doses.

Key assessment points: (1) identify all medicines and doses, (2) note when each medicine was started, (3) record when the symptoms began, (4) consider the patient's disease and other possible causes, (5) review what happened after the medicine was stopped or changed, and (6) document uncertainty rather than claiming that causality is proven.

3. Case Scenario 1 – Suspected Skin Rash with Amoxicillin

Patient profile: Fictional 28-year-old adult treated for a bacterial throat infection.
Medication history: Amoxicillin 500 mg orally, as prescribed, was started on Day 1. The patient reports no other newly started medicine in the same period.
Timeline and symptoms: On Day 3, the patient noticed an itchy red rash on the arms and trunk. The rash was not present before starting the antibiotic. The patient contacted a healthcare professional, who advised stopping and assessing the medicine. The rash began to improve over the following days.
Suspected drug: Amoxicillin.
Suspected reaction: Itchy skin rash (possible drug-related hypersensitivity reaction).
Assessment: The timing is compatible with a medicine-related reaction and improvement after stopping supports a possible association. However, infection-related rash, allergy history and other causes would need to be checked. **WHO-UMC category for this fictional information: Possible.**
Conclusion: Amoxicillin is the suspected medicine; the relationship is possible, not confirmed.

4. Case Scenario 2 – Suspected Gastrointestinal Reaction with Metformin

Patient profile: Fictional 46-year-old adult recently diagnosed with type 2 diabetes.
Medication history: Metformin 500 mg orally once daily with food was initiated by the treating clinician. No other new medicine is reported.
Timeline and symptoms: Loose stools, abdominal discomfort and nausea began within the first few days of treatment. The patient reports that these symptoms were not present before starting metformin. A clinician reviewed the symptoms and adjusted the treatment plan; the gastrointestinal symptoms subsequently improved.
Suspected drug: Metformin.
Suspected reaction: Diarrhoea, nausea and abdominal discomfort.

Assessment: The onset after starting metformin is temporally compatible with a suspected gastrointestinal ADR. Other explanations, such as dietary causes or gastrointestinal infection, should also be considered. **WHO-UMC category for this fictional information: Possible.**
Conclusion: Metformin is the suspected medicine. More clinical information would be needed to strengthen the causality assessment.

5. Case Scenario 3 – Suspected Dry Cough with Enalapril

Patient profile: Fictional 55-year-old adult receiving treatment for hypertension.
Medication history: Enalapril 5 mg orally once daily was started by the treating clinician. The patient had not reported a persistent dry cough before treatment.
Timeline and symptoms: A persistent dry cough developed approximately two weeks after starting enalapril. The clinician assessed the patient for other likely causes and changed the antihypertensive treatment. The cough gradually resolved after the change.
Suspected drug: Enalapril.
Suspected reaction: Persistent dry cough.
Assessment: The timing and improvement after the clinician-directed change support a drug relationship. If other causes are considered unlikely, the fictional case is consistent with **WHO-UMC Probable/Likely**. A deliberate re-challenge is not required and should not be attempted for this assignment.
Conclusion: Enalapril is the suspected medicine and the reaction is assessed as probable/likely in this fictional example.

6. Summary Table

Case	Suspected drug	Symptoms	Onset	WHO-UMC assessment
1	Amoxicillin	Itchy rash	Day 3	Possible
2	Metformin	Loose stools, nausea, abdominal discomfort	First few days	Possible
3	Enalapril	Persistent dry cough	About 2 weeks	Probable/Likely*

*The causality categories are practice assessments based only on the limited fictional details presented. A real case assessment requires complete and verified clinical information.

7. General Conclusion

ADR identification requires careful review of the medicine history, symptom description, onset, clinical condition, concomitant medicines and outcome after withdrawal or change of treatment. A time relationship alone does not prove that a medicine caused an event. The suspected relationship should be assessed systematically and documented with an appropriate level of certainty. Patients should not stop or change prescribed medicines without advice from a qualified healthcare professional.

8. References

1. Indian Pharmacopoeia Commission. Pharmacovigilance Programme of India (PvPI), Frequently Asked Questions. https://ipc.gov.in/PvPI/faq.html

2. Indian Pharmacopoeia Commission. PvPI Toolkit – Causality Assessment and WHO-UMC categories. https://ipc.gov.in/PvPI/toolkit.html

3. Indian Pharmacopoeia Commission. Guidance Document for Spontaneous Adverse Drug Reaction Reporting. https://www.ipc.gov.in/PvPI/pub/Guidance%20Document%20for%20spontaneous%20Adverse%20Drug%20Reaction%20Reporting.pdf

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